

Navigators Final Software Validation Report

Comprehensive Test Suite Coverage & Verification Audit

Navigators : Verification & Testing Protocol

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1. Test Architecture & Coverage Framework

The Navigators test suite employs an exhaustive multi-tier verification methodology covering mathematical algorithms, state machine transitions, REST API endpoints, security matrices, database constraints, and browser execution:

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graph TD
    TestRoot[Complete Test Harness (288 Automated Tests)] --> Pytest[pytest Backend Suite (274 Tests)]
    TestRoot --> NodeRunner[Node.js Edge Engine Suite (14 Tests)]

    Pytest --> UnitAlgo[Inertial Algorithms & EKF (57 Tests)]
    Pytest --> SecurityRBAC[RBAC, Auth & Security Matrix (78 Tests)]
    Pytest --> StateMachine[Lifecycle State Machines (22 Tests)]
    Pytest --> CRUD[Database CRUD & Repositories (65 Tests)]
    Pytest --> ML[Neural Models & Parity (52 Tests)]

    NodeRunner --> MotionTest[Vehicle Motion Detection (1 Test)]
    NodeRunner --> OfflineNav[Offline Navigation & LocalMap (13 Tests)]
```

2. Automated Test Suite Inventory (288 Tests)

2.1 Python Backend & Algorithm Suite (274 Tests via *pytest*)

Test File	Test Count	Subsystem Scoped	Key Invariants Verified
tests/test_ekf.py	15	Extended Kalman Filter	Covariance propagation, Joseph symmetry, measurement update validity.
tests/test_dead_reckoning.py	21	Dead Reckoning & NHC	Zero-velocity suppression, lateral drift gating, velocity integration.
tests/test_alignment.py	5	Triad Alignment	Gravity vector leveling, forward acceleration heading resolution, PCA fallback.
tests/test_nav_state.py	5	Navigation State Machine	Transitions between INIT, LOCKED, DEGRADED, DR, ZUPT, REACQ.
tests/test_security_matrix.py	8	Security Matrix	Rejections of unauthorized access across 7 roles and direct routes.
tests/test_authorization_service.py	10	Central Authorization	Permission evaluation, self-moderation prevention, account status checks.

<i>tests/test_database_rbac.py</i>	14	RBAC Database Engine	Role-permission junction joins, cascading revocations, privilege isolation.
<i>tests/test_ownership_rules.py</i>	2	IDOR & Ownership	Draft privacy protection, author-exclusive edit rights.
<i>tests/test_auth_sessions.py</i>	15	Session Security	SHA-256 token hashing, expiration, revocation, guest session bounds.
<i>tests/test_places_crud.py</i>	14	Canonical Places	Spatial queries, bounding box search, soft-deletion, place versioning.
<i>tests/test_state_machine.py</i>	10	State Machine Engine	Deterministic 5-stage contribution transitions; illegal jump rejection.
<i>tests/test_moderation.py</i>	8	Moderation API	Queue filtering, approval, rejection notes, audit trail creation.
<i>tests/test_datasets.py</i>	21	Telemetry Ingestion	Consent validation, continuity verification, session split assignment.
<i>tests/test_models.py</i>	17	TCN Architecture	Causal receptive field, forward tensor dimensions, parameter count.
<i>tests/test_model_registry.py</i>	7	Model Governance	Candidate promotion, deployment atomic swapping, instant rollback.
<i>tests/test_audit_logging.py</i>	6	Audit Ledger	Action formatting, actor recording, immutable ledger retrieval.
<i>tests/test_replay.py</i>	18	Replay Harness	Zero ground-truth leakage, ablation scoring, output trajectory schemas.
<i>tests/test_search.py</i>	5	Place Search	Text normalization, category filtering, proximity distance sorting.
<i>tests/test_offline_sync.py</i>	5	Offline Queue	Idempotency key deduplication, transaction integrity.
<i>tests/test_admin_dashboard.py</i>	4	Admin Endpoints	System telemetry consolidation, user role elevation.
<i>tests/test_internal_contributors.py</i>	11	Internal Role Access	Application submission, staff review, capability upgrade.
<i>tests/test_privacy.py</i>	8	Data Privacy	Coordinate obfuscation, user profile deletion, consent compliance.
<i>tests/test_sos.py</i>	5	Emergency Distress	Incident logging, distress broadcast, resolution status update.
<i>tests/test_ui_integration.py</i>	5	UI Integration	Contract matching between backend models and frontend client state.

2.2 Node.js Edge Engine Suite (14 Tests via *node --test*)

Test File	Test Count	Key Invariants Verified
tests/vehicle_motion_detection.test.cjs	1	Complete motion transition cycle: Stationary $\rightarrow$ Moving (GNSS) $\rightarrow$ Outage $\rightarrow$ ZUPT Stop $\rightarrow$ Motion Recovery.
tests/offline_navigation.test.cjs	13	Coordinate frame round-trips, long road segment projections, causal IMU filtering, zero GPS origin invention, anti-jump reacquisition capping ($\leq 2\text{m}$).

3. Separation of Automated Software Tests vs. Physical Hardware Tests

> [!IMPORTANT]

> **Strict Operational Distinction:**

- > - **Automated Software Validation (PASSED: 288/288):** Validates that all mathematical equations, EKF routines, TCN models, state transitions, security barriers, and offline engines execute correctly against digital inputs and synthetic scenarios.
- > - **Physical Hardware Validation (BLOCKED / PENDING ROAD TRIALS):** Validates physical smartphone MEMS noise under real vehicle vibration, real thermal dissipation inside a hot car cabin, and real satellite attenuation in physical road tunnels. This remains unmeasured until field road testing is conducted.

Developed by Navigators